

Whole-Home Back-up solution Via External ATS



Introduction

In general, there are two mainstream solutions for household loads distribution on the market.

(1)Partial-Home Back-up solution

In this solution, users' household loads will be separated into Common Loads and EPS Loads, only EPS Loads can be powered during EPS mode.

(2)Whole-Home Back-up solution

In this solution, users' whole household loads will be able to be powered during EPS mode if power supply is enough.

These two solutions are well related to the transferring methods between On-grid & EPS status.

External ATS (Automatic Transfer Switching Device) is an equipment installed outside of the inverter, which can automatically transfer between On-grid & EPS status. That is to say, if users want to power their whole household loads during off-grid, an External ATS device is necessary for Hybrid Inverter.

Moreover, **[External ATS]** also means a commission on the LCD settings of inverters.

The following table presents all accessories and their corresponding features.

Accessories	Features
X1/ X3-EPS BOX	With a DPDT equipment contained; For whole-house power distribution;
X3-EPS BOX G2	Dual power(motor) supply version of the

	distribution box for whole-Home back-up solutions;
X3-EPS Parallel BOX	First generation inverter parallel box, product maximum parallel 50kW;
X3-EPS Parallel BOX G2	Second generation inverter parallel box, the product has 3 versions, 60kW, 150kW, 300kW, the product is mainly used for whole-house power distribution;
X1/X3 Matebox Basic	All in one installation for Partial-Home Back-up solutions ;
X1/X3 Matebox Advanced	Built-in DPDT, all in one installation, suitable for whole-Home back-up solutions ;
X1/X3 Matebox G2 X1/X3 Matebox G2-M	G2 is the version with DPDT , G2-M is the version with motor ;

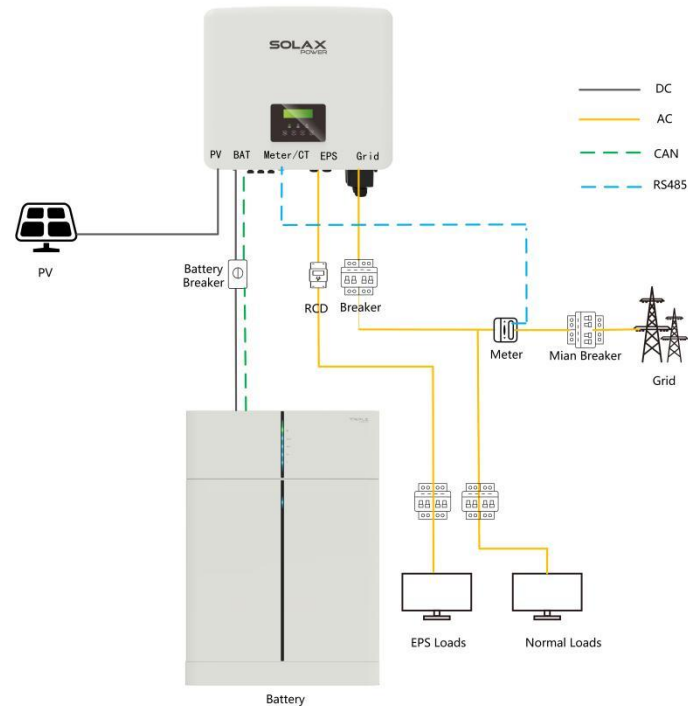
Partial-Home Back-up & Whole-Home Back-up

1. Partial-Home Back-up

As for Partial-Home Back-up, built-in transferring switches will be used to change the output port when the inverter enter into EPS mode causing by power outage. And the users' household loads have to be separated into Common Loads and EPS

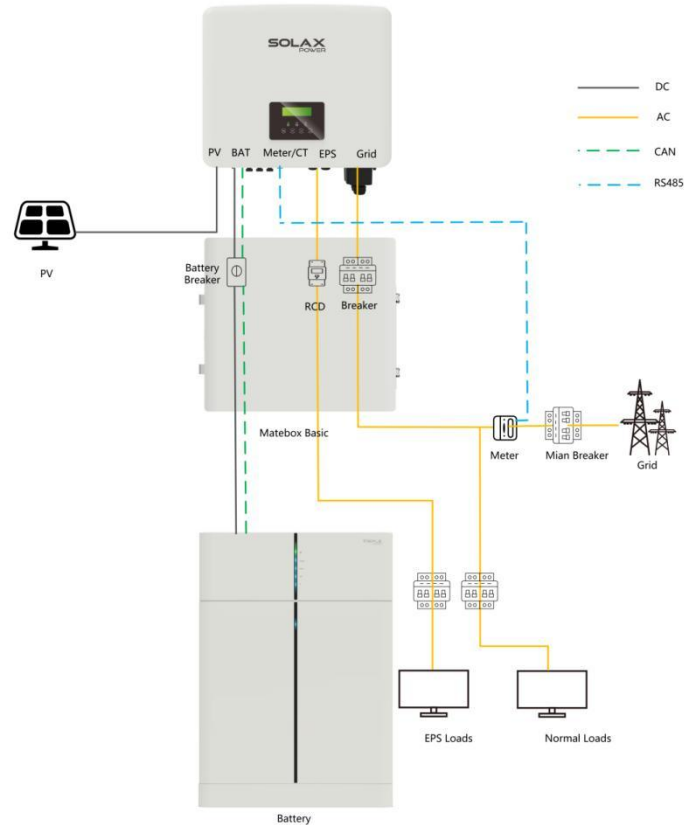
Loads. Only EPS Loads will be powered during EPS mode such as lights and refrigerators . The system diagram is shown below.

① Normal System



Note: Need to disable **[External ATS]**, without Matebox (need to install external protection devices).

② With Matebox Basic

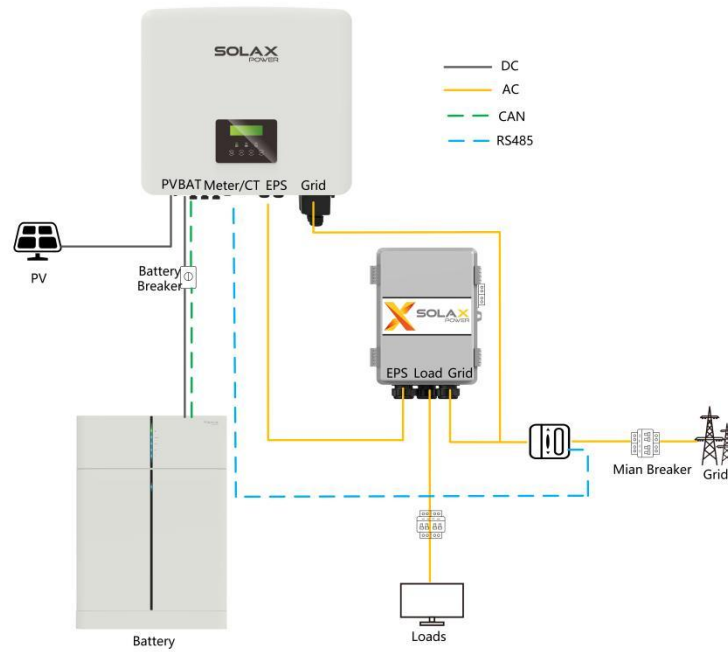


Note: Need to disable **[External ATS]**, with Matebox Basic, with T30 Batteries to form All-in-one Solution.

2. Whole-Home Back-up

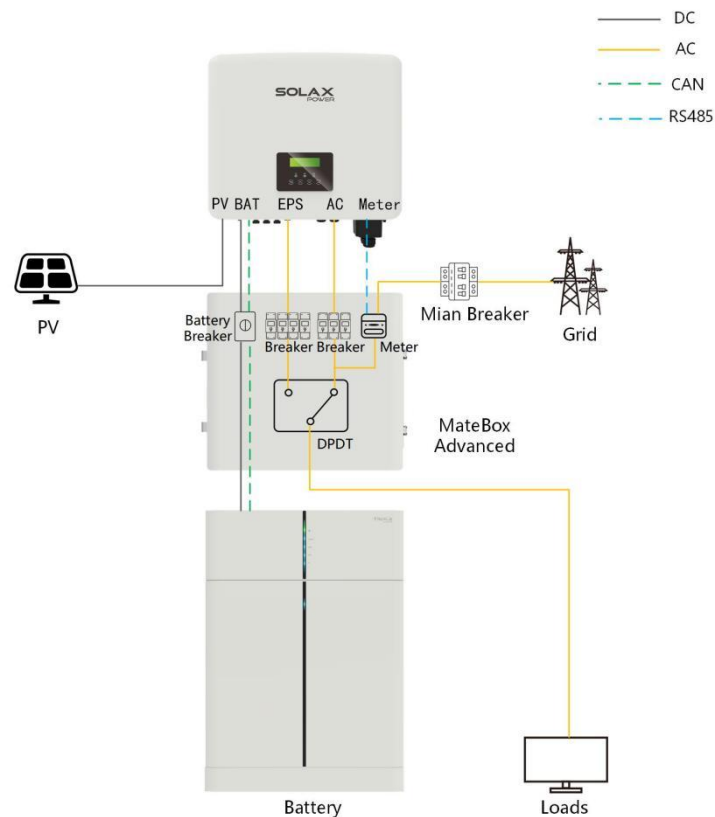
Whole-Home Back-up can be achieved by external switching devices for Hybrid inverters and the inverters should have the External ATS function, in which users' whole household loads can be powered during off-grid. A Whole-Home Back-up solution is shown below.

① With EPS Box



Note: Need to Enable [External ATS], the load power should be no more than the maximum EPS output power.

② With Matebox Advanced



Notes:

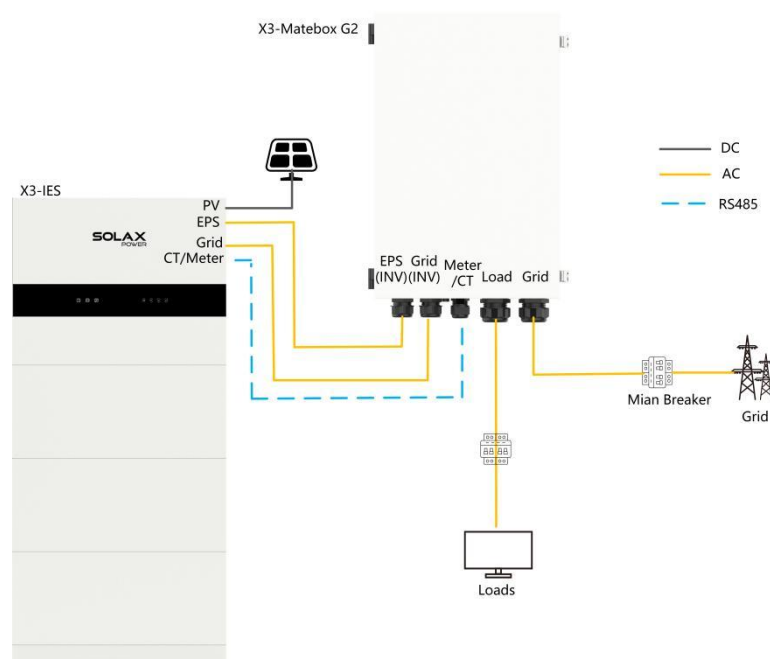
1. Need to Enable **[External ATS]**

2. The total power of the loads run simultaneously should be no more than EPS output power.

3. Different transferring devices cannot be applied simultaneously, for example, a Matebox Advanced shall not exist in parallel solutions with EPS Parallel Box.

4. For X3-Ultra, the whole home loads can be connected to EPS port to realize the whole-home backup by the internal transferring switch.

③ **With Matebox G2**



Notes:

1. Need to Enable **[External ATS]**

2. Matebox G2 supports connection to various inverters. We take IES as an example to describe how to connect the inverter. The specific compatibility tables are at the bottom of the document

3. The Matebox G2 adopts a mechanical interlock (DPDT) contactor, which switches time can be within 100ms, but there will be sound at the moment of suction. The Matebox G2-M adopts a dual power motor switching contactor, which switches time is within 2s, but the sound is very small at the moment of suction.

Setting path:

Main → Settings → Advance Setting → External ATS

This setting is **Disable** by default. If any **External ATS** device (EPS Box, EPS Parallel Box , Matebox Advanced, Matebox G2) mentioned above is used, select **Enable** to enable this function.

Compatibility

Accessory	Hybrid Inverter
X1-EPS BOX	X1-Hybrid-G4-D
X3-EPS BOX	X3-Hybrid-G4-M X3-Hybrid-G4-D X3-IES X3-IES-P X3-Hybrid G4 PRO X3-ULTRA
X3-EPS BOX G2	X3-Hybrid-G4-M X3-Hybrid-G4-D X3-IES

	X3-IES-P X3-Hybrid G4 PRO
X3-PBOX-60kW-G2	X3-Hybrid-G4-M X3-Hybrid-G4-D X3-IES X3-IES-P X3-Hybrid G4 PRO
X3-PBOX-150kW-G2.1	X3-Hybrid-G4-M X3-Hybrid-G4-D X3-IES X3-IES-P X3-Hybrid G4 PRO X3-ULTRA
X3-PBOX-300kW-G2	X3-ULTRA X3-AELIO AELIO P50B100/ AELIO P60B100 AELIO P50B200/ AELIO P60B200
X1/X3 Matebox Basic V1.3	X1-Hybrid-G4-M X3-Hybrid-G4-M
X1/X3 Matebox Basic V1.4	X1-Hybrid-G4-D X1-Hybrid-G4-M

	X3-Hybrid-G4-D X3-Hybrid-G4-M
X1/X3 Matebox Advanced V1.3	X1-Hybrid-G4-M X3-Hybrid-G4-M
X1/X3 Matebox Advanced V1.4	X1-Hybrid-G4-D X1-Hybrid-G4-M X3-Hybrid-G4-D X3-Hybrid-G4-M
X1/X3 Matebox G2 X1/X3 Matebox G2-M	X1-IES X3-Hybrid G4-D X3-IES X3-IES-P X3-Hybrid G4 PRO